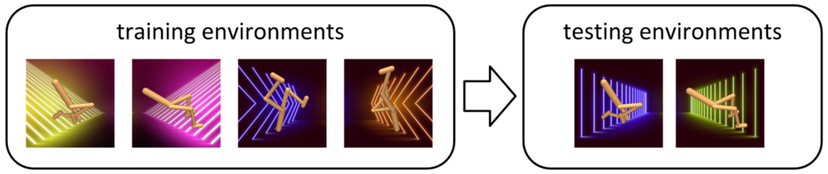
Domain Adversarial Reinforcement Learning

Bonnie Li Vincent Francois-Lavet Thang Doan Joelle Pineau
McGill University Mila VU Amsterdam

Overview

We consider the problem of generalization in reinforcement learning where visual aspects of the observations might differ. For example, when there are different backgrounds.

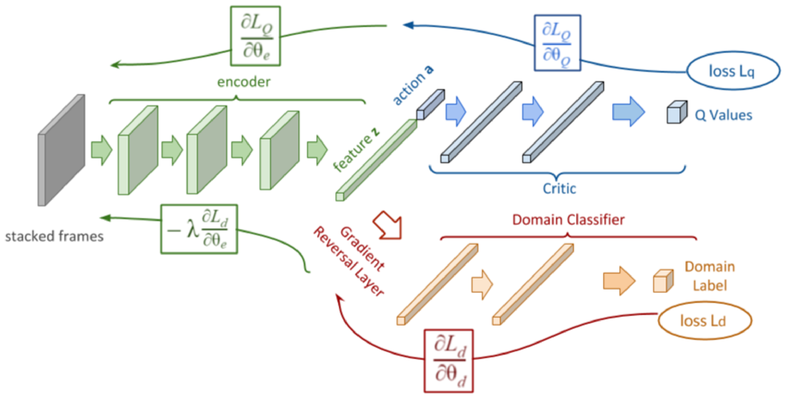
We assume that our agent has access to only a few of the MDPs from the block MDP distribution during training. The performance of the agent is then reported on new unknown test domains taken from the distribution (e.g. unseen backgrounds).



DARL

We introduce DARL: Domain Adversarial Reinforcement Learning, which aims to improve generalization to all visual domains with the same task dynamics.

DARL aims to learn representations that
1. capture task relevant features in source domains
2. are invariant with respect to domain distribution shift

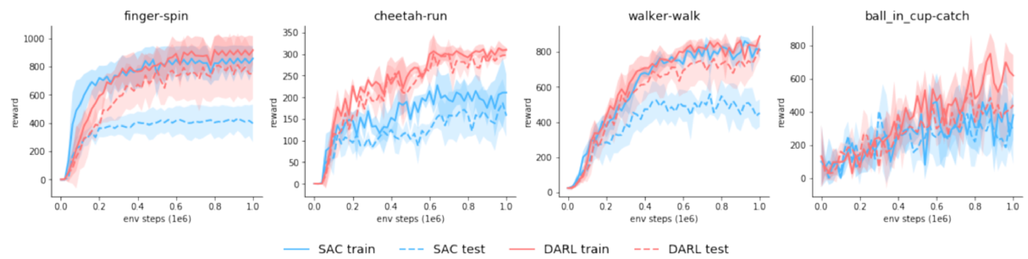


DARL builds a domain adversarial module along with the critic and encoder of SAC.
Adversarial Discriminator: the domain classifier, tries to predict which domain the feature representations come from.
Gradient Reversal Layer: connects the domain classifier and the encoder, and reverses the gradient during backpropagation. By reversing the gradient, the encoder is trained to fool the domain classifier.

By directly minimizing the domain classification loss via backpropagation, we train the domain classifier to distinguish the background class from state features, and simultaneously train the encoder to produce state features that **maximizes** the classification error (by taking the reversed gradient), therefore encouraging invariant features that are indistinguishable across domains.

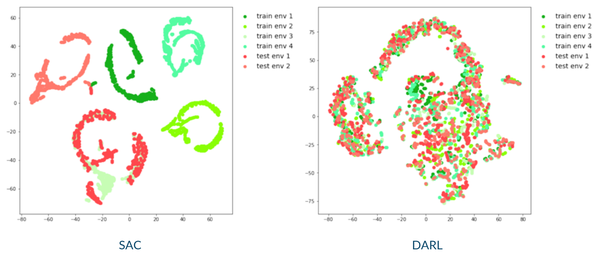
Experiments

Comparison Against Baselines



We train the agent in four stationary environments, each with a different background, and directly evaluate the agent on two testing environments with unseen backgrounds. DARL outperforms SAC on testing environments across all four tasks.

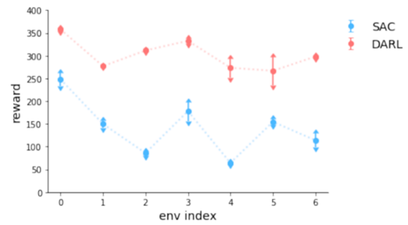
Visualizing Latent Representations



We collect one trajectory in each training and testing environment, and visualize the latent representations of the collected observations via a t-SNE plot.

While SAC learns different embeddings for different backgrounds, DARL outputs the same embedding independently of the background.

Generalization to Natural Video Backgrounds



Can DARL zero-shot generalize to natural video backgrounds?
We take the models trained previously on stationary backgrounds, and directly evaluate them on natural video environments.

DARL is more robust to visual variations and more stable in performance across all environments,

project page: bonniesjli.github.io/darl | email: bonnie.li@mail.mcgill.ca